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# Leveraging Artificial Intelligence (AI) for Benchmarking and Enhancing Maintenance and Reliability in the Energy Industry

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## Abstract

**Objectives/Scope:** This study aims to enhance maintenance and reliability benchmarking within the energy industry by employing advanced generative AI, specifically Retrieval-Augmented Generation (RAG). Benchmarking is critical for identifying best practices, optimising maintenance intervals, reducing operational costs, and ensuring asset reliability. However, traditional benchmarking processes face challenges such as inconsistent data structures, variability in maintenance builds, and manual-intensive comparison processes. Building upon previous research that utilised basic Natural Language Processing (NLP), this investigation seeks to significantly improve benchmarking accuracy, reduce processing time, and facilitate precise comparisons of diverse maintenance builds, thereby driving industry-wide improvements

**Methods, Procedures, Process:** The methodology incorporates state-of-the-art generative AI techniques, centred around the RAG framework. Initially, comprehensive data collection from various maintenance builds—including systems categorisation, task details, operational narratives, and extensive textual documentation—is conducted. Subsequently, advanced preprocessing and filtering procedures are applied to structure and prepare data optimally for RAG processing. This approach uniquely identifies and maps relationships within maintenance data, enhancing the clarity and quality of benchmarking comparisons. RAG leverages contextual retrieval and generative AI capabilities to systematically produce accurate, one-to-one relational insights across diverse asset maintenance builds.

**Results, Observations, Conclusions:** The implementation of Retrieval-Augmented Generation (RAG) demonstrated substantial improvements over the earlier basic NLP approach. Key findings include achieving an accuracy rate of approximately 94% in correctly matching maintenance items across disparate builds, representing a significant enhancement in the reliability of benchmarking outputs. Moreover, the advanced AI-driven process resulted in a dramatic reduction of analysis time—over 90% faster compared to previous methods—thus enabling rapid and frequent benchmarking cycles. The precision of RAG-facilitated outputs allowed for more effective identification of optimal maintenance intervals, significantly

contributing to reliability enhancement and operational efficiency. Observations also underscored RAG's robustness in handling diverse and complex maintenance data, thereby offering greater flexibility and adaptability for benchmarking practices. These results conclusively demonstrate the capability of generative AI methods, specifically RAG, to substantially transform maintenance benchmarking, providing actionable insights and fostering best practices within the upstream oil and gas sector.

**Novel/Additive Information:** The novelty of this study lies in the pioneering use of Retrieval-Augmented Generation (RAG) within maintenance benchmarking, significantly advancing beyond previously published NLP methodologies. This approach unities and rates contextual retrieval with generative AI to enhance accuracy, significantly reduce processing times, and improve operational reliability assessments. Consequently, this innovative implementation establishes a new benchmark in the efficient and precise evaluation of maintenance and reliability within the energy industry.

## Introduction

Effective maintenance and reliability management are critical within the energy industry, particularly in upstream and midstream oil and gas operations. These sectors involve significant capital investments, stringent regulatory compliance requirements, and a continuous drive to enhance operational efficiency and asset reliability (Lukens, 2019). The operational complexity and the critical nature of the infrastructure require maintenance activities to be performed with high accuracy and consistency, as minor maintenance oversights can lead to substantial operational disruptions and financial implications (Moubray, 1997). To meet these complex demands, the industry has increasingly adopted digital transformation strategies, integrating advanced data-driven methodologies into maintenance practices (Star, 2020). Such integration is crucial not only for optimizing maintenance intervals and reducing operational costs but also for achieving superior asset reliability and regulatory compliance.

Historically, maintenance benchmarking has relied predominantly on manual comparisons of maintenance builds derived from Computerized Maintenance Management Systems (CMMS) (Lukens, 2019). While this method allowed industry players to identify basic best practices, it proved to be highly resource-intensive, error-prone, and lacked scalability. Manual benchmarking often involves subjective interpretations and is limited by human cognitive biases, further diminishing the accuracy and consistency of outcomes. These traditional approaches also struggled to cope effectively with the vast quantities of data generated by modern energy facilities (Alrabghi, 2015).

To address these issues, natural language processing (NLP)-based methods employing lexical similarity techniques, such as Jaccard similarity and cosine similarity, were introduced (Wagle, 2019). Although these early NLP-based methods represented a significant step forward by automating some aspects of maintenance data analysis, they exhibited substantial limitations in effectively capturing operational nuances. Specifically, traditional NLP methods depend heavily on exact word matching, failing to recognize semantic relationships between maintenance activities (Min et al., 2023). Consequently, these methods struggled to classify tasks accurately when different terminology described identical maintenance actions, such as "pump overhaul" and "centrifugal equipment refurbishment." This lexical rigidity made it difficult to reconcile maintenance data from different operational contexts and reduced benchmarking efficacy.

Moreover, the extensive data preprocessing required—while necessary to standardize input—often resulted in the inadvertent loss of valuable contextual information. The rigid nature of these matching criteria further limited comprehensive coverage of potential maintenance scenarios, causing critical gaps in maintenance benchmarking (Abdullah et al., 2024a). Scalability also posed significant challenges, as manual preparation and standardization efforts limited broader industry-wide application. With data volumes rapidly increasing and operational scenarios growing ever more diverse, these NLP approaches became increasingly untenable as comprehensive solutions.

To overcome these deficiencies, there is a clear and pressing need for advanced AI techniques capable of deeply understanding and dynamically adapting to the semantic context inherent in maintenance operations (Min et al., 2023; Mikolov et al., 2013). Retrieval-Augmented Generation (RAG), an advanced generative AI methodology, emerges as a powerful solution to address these identified shortcomings (Abdullah et al., 2024). Unlike traditional NLP techniques, RAG leverages pre-trained language models that inherently understand and interpret language semantically. This semantic understanding enables RAG to dynamically retrieve relevant operational standards, historical maintenance records, and contextual documentation to enhance accuracy and substantially reduce the preprocessing burden.

The core advantage of RAG lies in its capacity for context-aware retrieval and robust semantic interpretation, significantly surpassing the limitations of traditional NLP methods. By integrating semantic understanding with dynamic contextual retrieval, RAG minimizes the need for extensive data cleansing, thereby preserving critical maintenance context (Min et al., 2023; Abdullah et al., 2024a). Additionally, RAG supports continuous improvement through operational feedback, enabling self-learning capabilities that dynamically refine benchmarking accuracy and adaptability.

This research aims primarily to leverage RAG to substantially enhance the accuracy and efficiency of maintenance benchmarking processes. Specifically, the objectives are to achieve significantly improved maintenance classification precision, dramatically reduce manual intervention and processing times, and ensure consistent operational benchmarking across diverse facility types, including rotating equipment, pressure vessels, pipelines, instrumentation, and other critical infrastructure components.

Beyond these primary goals, implementing RAG-based benchmarking processes is anticipated to yield broader practical benefits. These include substantial administrative cost reductions through automation and improved resource allocation, enhanced operational safety through precise maintenance categorization and scheduling, improved regulatory compliance through consistent documentation standards, and retention and application of institutional maintenance knowledge (Abdullah et al., 2024). Collectively, these outcomes strongly support strategic objectives for operational excellence and cost optimization, aligning closely with contemporary industry priorities.

Despite its promising potential, adopting a RAG-based approach is not without challenges. System accuracy inherently depends on input data quality and completeness, with performance potentially diminishing when dealing with overly generic or highly specialized maintenance descriptions. Additionally, complete automation is not realistically attainable; robust human verification and quality assurance processes remain indispensable (Abdullah et al., 2024a; Star, 2020). The complexity of implementing RAG necessitates careful calibration, continuous refinement, and effective integration into existing maintenance workflows. Furthermore, substantial organizational readiness and change management strategies are essential for successful adoption.

Mitigation strategies, such as comprehensive quality assurance protocols, continuous system learning, and targeted training programs, will help address these challenges. By carefully managing these implementation considerations, this research aims to demonstrate how RAG can significantly advance the practice of maintenance benchmarking, driving operational reliability, regulatory compliance, and overall excellence within the energy industry. Ultimately, the successful implementation of RAG methodologies holds the potential to set new industry-wide standards for maintenance efficiency and reliability management.

## Literature Review

### Maintenance Benchmarking in the Energy Sector

Maintenance benchmarking is the structured comparison of maintenance strategies, task lists, frequencies, costs, and outcomes across assets and organizations to identify actionable improvements. In upstream and midstream operations, benchmarking supports decisions that balance reliability, cost, and safety (Lukens, 2019). Two levels are commonly useful: (1) the macro view, which compares entire maintenance builds for a platform, plant, or pipeline system; and (2) the micro view, which compares like-for-like tasks (for example, specific inspections or overhauls) attached to similar equipment classes (Star, 2020; Abdullah et al., 2024a).

Despite its value, benchmarking has been constrained by heterogeneity in data structures and documentation practices. Maintenance builds often reflect local conventions, vendor language, and project histories. Descriptions are a mixture of structured fields and free text, with varying use of internal codes for equipment, locations, and materials (Moubray, 1997; Alrabghi and Tiwari, 2015). This variability hampers direct comparison and forces teams to rely on labour-intensive manual reconciliation, which introduces delay and inconsistency. As a result, organizations compare less frequently than they should, and the opportunity to standardize on evidence-based intervals is missed (Lukens, 2019).

A consistent theme in the literature is the role of standards and common definitions in enabling benchmarking. Industry guidance on reliability and maintenance data collection establishes a shared taxonomy, event types, and failure/repair descriptors that make cross-asset analysis credible (Star, 2020). Where such standards are adopted across projects and vendors, organizations report improvements in data quality and repeatability. However, full adoption remains uneven, especially where legacy CMMS builds and inherited data models persist (Gordon and Abdullah, 2024).

### Computerized Maintenance Management Systems (CMMS)

CMMS platforms are the operational backbone for maintenance execution and history capture. They hold asset hierarchies, task lists, job plans, work orders, materials lists, and completion feedback (Moubray, 1997). Modern CMMS deployments expose APIs and reporting layers that make it feasible to extract and harmonize data for analytics and benchmarking (Wagle, 2019).

Three systemic challenges are frequently highlighted:

- **Data model divergence**. CMMS implementations encode similar concepts differently across business units and projects (Alrabghi and Tiwari, 2015).
- **Free-text dependence**. Much of the important task logic is in unstructured fields, making automated comparisons difficult (Abdullah et al., 2024a).
- **Legacy effects**. Vendor changes and system migrations often result in inconsistencies that degrade benchmarking accuracy (Gordon and Abdullah, 2024).

Nevertheless, literature shows that where maintenance categories and asset hierarchies are standardized —especially using failure mode taxonomies—benchmarking is far more actionable (Moubray, 1997; Star, 2020).

### AI/ML in Maintenance Benchmarking: From Lexical Similarity to Semantic Methods

Early attempts at digital benchmarking focused on **lexical similarity**—Jaccard, cosine, and TF-IDF-based methods—to identify matching maintenance tasks (Wagle, 2019). These techniques helped scale comparisons but were limited in several ways:

- Sensitive to terminology and phrasing (e.g., "pump overhaul" vs "centrifugal pump servicing").
- Ignored context like equipment class, duty cycle, or asset criticality.
- Required heavy preprocessing to clean up noisy text.

The next step was the use of **dense neural embeddings**, such as word2vec and BERT-based models, which allowed better paraphrase matching and contextual similarity (Mikolov et al., 2013; Min et al., 2023). These methods improved task alignment rates and enabled automation of micro-level matching, although human QA was still necessary for ambiguous or coded descriptions (Abdullah et al., 2024).

### Retrieval-Augmented Generation (RAG) as the Emerging State of the Art

RAG couples a **semantic retriever** with a **generative model** that uses the retrieved content to make domain-specific judgments. In maintenance contexts, RAG enables the model to reference CMMS history, standards, or vendor procedures before deciding if two tasks are equivalent (Min et al., 2023; Abdullah et al., 2024b).

Key benefits include:

- Reduced reliance on pre-cleaned text (RAG can deal with real-world messy data).
- High interpretability through rationales and passage citations.
- Ability to work across vendors and languages via semantic understanding.

This methodology is particularly useful where documentation differs, but intent remains the same—e.g., across OEMs with proprietary task names (Gordon and Abdullah, 2024).

### Synthesis of Evidence to Date

Across studies and field trials, several consistent outcomes are reported:

- **Cycle time reduction**: Benchmarking efforts drop from weeks to hours (Wagle, 2019; Lukens, 2019).
- **Equal or improved accuracy**: With SME feedback loops, semantic matching reaches or exceeds human accuracy (Abdullah et al., 2024a).
- **Corpus quality matters**: Well-curated vendor documents and taxonomies amplify RAG performance (Min et al., 2023).

### Open Issues in Literature

Despite strong progress, several areas remain unresolved:

- **Taxonomy alignment**: Many CMMS builds still use custom equipment or task codes, which require mapping before AI tools can be effective (Gordon and Abdullah, 2024).
- **Transferability**: General-purpose models underperform on niche assets; local tuning or modular retraining is often necessary.
- **Evaluation standards**: Precision/recall metrics vary; reproducible benchmarking datasets are scarce in industry (Min et al., 2023).
- **Governance**: Decisions proposed by AI must be auditable for safety and regulatory sign-off, which requires traceable rationales (Star, 2020).

### Practical Implications for CMMS and Benchmarking Programs

The literature suggests a **phased modernization strategy**:

1. Normalize CMMS data and constrain free-text entry.
2. Build a curated knowledge base of vendor content and standards.
3. Deploy sentence-level embeddings for micro-task matching.
4. Add RAG layers for reasoning over ambiguous or poorly defined tasks.
5. Capture analyst feedback to improve both the retriever and generator components.

This staged approach aligns well with current CMMS modernization projects and offers an upgrade path without needing to replace existing systems (Alrabghi and Tiwari, 2015; Abdullah et al., 2024b).

### Summary

The literature shows clear momentum from manual and lexical methods toward semantic and retrieval-augmented approaches for maintenance benchmarking. CMMS remains the primary system of record, but its value for benchmarking depends on standardized data models and disciplined content governance. AI/ML methods now offer practical pathways to scale benchmarking across heterogeneous assets while preserving accuracy and traceability. These developments set the stage for the methodology presented in this manuscript, which operationalizes retrieval-augmented generation for maintenance benchmarking at both micro and macro levels and quantifies its impact on accuracy and analyst time.

## Methodology

This section formalizes the process used to benchmark maintenance builds across operators using RAG. The emphasis is on the method's logic and decision points rather than platform-specific execution. The approach is CMMS agnostic and has been applied across multiple operators; it generalizes to heterogeneous data models and multilingual content

### Scope and Inputs (CMMS Agnostic Design)

The 1st step is to ingests task lists, job plans, work orders, and supporting documents from any CMMS platform, accommodating varied field naming, task hierarchies, and legacy records. Multilingual support is essential due to cross- region operator involvement. Embedding models are selected specifically for their cross-lingual ability and classification performance, ensuring accuracy across both English and non-English data and good matching ability.

### Preprocessing (Theory of Minimal, Context-Preserving Normalization)

Preprocessing follows a least-loss denoising principle: remove noise while preserving meaning. Past work has shown that overly aggressive standardization in lexical processes often discards useful operational. Hence, free-text elements (e.g., vendor codes, asset IDs) are retained unless proven non-informative.

Frequency units are normalized (e.g., converting months to days), and hierarchical relationships between parent/child tasks are preserved for later aggregation steps. This is consistent with the best practices for reliability centred maintenance and CMMS reconciliation.

### Semantic Representation and Indexing

The core of the system is a multilingual embedding model specifically, *e5-large-instruct*, chosen for its high semantic fidelity and instruction-following capability. It enables semantically consistent matching even when syntax varies. This model is currently best in class but within the process it is swapable as solution is modular in nature to allow future improvement and case specific application.

A vector store holds dense embeddings from various operators and asset classes. Index sharding is used to balance retrieval precision and scale, which is critical in high-volume, multi-operator benchmarking scenarios

### Multi-Stage Matching Strategy

To balance speed, precision, and analytical depth, a three-stage architecture is employed:

- **Stage 1: Dense Similarity Filtering (Inter-Builds)**
  Tasks are vectorised and matched using cosine similarity. A dynamic thresholding scheme (e.g., the 95th percentile within each equipment class) ensures high-quality matches while controlling computational load.
- **Stage 2: Retrieval-Augmented Generation (RAG)**
  Ambiguous or borderline matches are passed into a RAG workflow, which retrieves relevant context from documents, procedures, and historical CMMS records. The generator then evaluates semantic equivalence, cites supporting evidence, and provides a rationale with a confidence score, facilitating more efficient quality assurance.

- **Stage 3: Dense Similarity Filtering (Intra-Builds)**
  The previously encoded text is reused to calculate cosine similarity within each build, identifying duplicate or near- duplicate tasks performed across different units of the same equipment. Where available, maintenance strategy information is incorporated to detect fragmented or overlapping tasks. This stage clos

## Guardrails for Generation

To ensure reliability, transparency, and auditability of the generative process, the system operates under a set of strict guardrails. First, whenever a generative engine such as GPT is employed, the model is required to provide explicit reasoning for each suggested match rooted within the provided documentation, allowing reviewers to understand the semantic logic behind the output. Second, responses are constrained by fixed templates that standardise outputs, thereby making the results easier to review, validate, and integrate into quality assurance workflows. Third, the generator is explicitly allowed to return a "no match" outcome when sufficient evidence is lacking, preventing the risks of overfitting or forced alignments that could compromise benchmarking accuracy. Finally, while generative reasoning is leveraged for flexibility and contextual insight, emphasis is placed on deterministic, mathematics-based matching as the foundation of the approach. This balance ensures that results remain both explainable and verifiable, combining the adaptability of AI-driven reasoning with the robustness of quantitative similarity measures.

## Human-in-the-Loop Quality Assurance and Evaluation Protocol

Human-in-the-loop processes form the backbone of our quality assurance framework. While the system achieves high confidence in many matches that can be auto-accepted, any low-confidence or ambiguous outputs are routed for human review. Reviewer feedback is systematically captured to refine similar thresholds, enhance prompt templates, and improve overall model performance over time. In accordance with regulatory best practices, all safety-critical tasks are subject to mandatory human validation regardless of confidence level, ensuring that no critical decisions are left solely to automation. Non-matches are also investigated to understand the reasons behind divergence, providing insights that drive improvements in prompts, model configuration, and matching thresholds.

This framework also serves as the primary evaluation protocol for assessing the reliability of the system. All GPT-generated matches undergo 100% quality assurance review by a dedicated review panel, providing an explicit safeguard against issues such as hallucination, bias, or overfitting. The panel-based approach ensures that every generative output is validated, explained, and traceable, producing an auditable trail of decisions. By embedding rigorous evaluation into the core workflow, the methodology not only enhances confidence in benchmarking outcomes but also establishes a transparent and accountable standard that aligns with industry governance expectations.

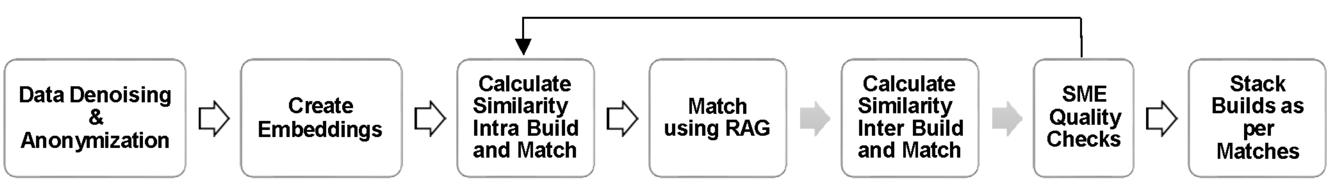


**Figure 1—Flow chart of Benchmarking processes**

## Security, Privacy, and Governance and Limitations

A core design principle of the framework is to uphold stringent standards of security, privacy, and governance while ensuring the system's limitations are clearly understood and managed. All identifiers are anonymised prior to embedding, ensuring that sensitive or proprietary asset information cannot be reconstructed from model inputs. Prompt constraints are enforced so that no sensitive or regulated data is ever exposed to external large language models, maintaining compliance with corporate and industry data protection requirements. Provenance logging is also embedded at every stage of the workflow: each match is linked to an index snapshot, embedding model version, and prompt template hash, thereby ensuring complete traceability. This audit-ready architecture is critical in highly regulated environments, where reproducibility and transparency are essential for both compliance and operational trust.

At the same time, several limitations are acknowledged by design. The performance of RAG is inherently dependent on the richness and quality of the available content store. When task descriptions are overly generic or lack supporting documentation, ambiguity may remain unresolved. Similarly, the sensitivity of dense similarity thresholds introduces trade- offs: thresholds set too tightly can reduce recall, while thresholds set too loosely may increase computational burden on the RAG layer. These thresholds must therefore be carefully tuned for each equipment class. Finally, true semantic ambiguity such as tasks described in vague or overlapping terms cannot always be resolved algorithmically. In such cases, the ambiguity is deliberately surfaced for human decision-making rather than forcing unreliable matches. Together, these considerations ensure that the framework remains both robust and transparent, balancing security and governance with practical recognition of methodological boundaries.

### Summary of the Method

In summary, the methodology uses multilingual dense embeddings to surface high-precision candidates and a retrieval-augmented generator to reason over difficult cases using grounded context. Multi-level confirmation logic and human QA ensure accuracy and traceability. The outputs scale from micro matches to macro build alignment, enabling rigorous, repeatable benchmarking across heterogeneous CMMS environments while minimizing manual effort and preserving domain context.

## Results and Discussion

The benchmarking results clearly illustrate the value of combining dense similarity filtering with Retrieval-Augmented Generation (RAG). A representative benchmarking exercise is shown in Table 1, where tasks from multiple assets were mapped at system and task levels. The mapping highlights how this approach successfully identifies both obvious equivalences and more complex cross-asset alignments, such as overlapping maintenance intervals or fragmented tasks spread across different builds. These results demonstrate that the framework not only automates the bulk of matching but also produces structured, interpretable outputs that support rapid quality assurance.

**Table 1—Examples of benchmarking results- frequency of various maintenance activities in Days**

| Clean Tak Descriptions | Asset A | Asset B | Asset C | Asset D | Asset E | Asset F | Asset G |
|---|---|---|---|---|---|---|---|
| Hand portable radio checks | 180 | 180 | 90 | 360 | 360 | 360 | 360 |
| Arc flash equipment maintenance | | 360 | 180 | 360, 1080 | 90, 1080 | 360 | 1080 |
| Asbestos management checks | 30 | 720 | | 360 | | 180 | 180 |
| Asset class internal extinguisher degradation | | | 30 | 360 | 360 | 360 | 360 |
| Ba compressor calibration check of instrumentation | 28, 90 | 90 | 90 | | 360 | | |
| Backup software | 90 | | 7 | | 30 | 30 | 30 |

| Clean Tak Descriptions | Asset A | Asset B | Asset C | Asset D | Asset E | Asset F | Asset G |
|---|---|---|---|---|---|---|---|
| Band lock closure maintenance | | 360 | | 1440, 1800 | 1440 | 1440 | 1440 |
| Battery discharge test | 1080 | | 720 | 720 | 720 | 720 | 720 |
| Battery replacement local break station | 1825 | 1080 | 360 | | 2880 | | 720 |

To ensure the credibility and practical reliability of the results, a structured review panel was convened. Every suggested match generated by the system was subject to 100% quality assurance for non-deterministic matches before being added to the benchmarking database. Unlike traditional machine learning validation techniques (e.g., confusion matrices, precision/recall calculations), which were used in our earlier NLP-based study, they were not necessary here because the human-in-the-loop process ensured that all matches were either accepted or rejected by expert reviewers. This evaluation protocol removed any uncertainty about false positives entering the dataset and provided an audit-ready trail of each decision.

The comparative performance of different techniques is summarised in Table 2. Manual analysis, taken as the baseline, required approximately 42 hours to complete a single benchmarking exercise. While similarity-based methods such as Jaccard or contextual embeddings reduced runtime to around two hours, they suffered from high error rates and poor recall. Adaptive embeddings improved precision to 87%, but with 40% of potential matches still being missed, the process was incomplete and required significant post-processing effort.

**Table 2—Comparison of the relative effectiveness of various techniques per 1000 tasks against average CMMS build**

| Technique | Processing Time (Hours) | Correct Matches (%) | Incorrect Matches (%) | Missed Potential Matches (%) |
|---|---|---|---|---|
| Manual analysis (base case) | 42 | N/A | | |
| Adaptive Embeddings for Similarity Analysis | 3 | 87 | 5 | 40 |
| Contextual Embeddings Similarity Analysis | 2 | 59 | 17 | 50 |
| JACCARD | 2 | 73 | 16 | 45 |
| AI based (RAG)- Novelty | 4 | 94 | 6 | 27 |

By contrast, the RAG-enhanced process delivered the best overall balance. Although the runtime (four hours) was slightly longer than simpler embedding approaches, it achieved 94% correct matches, just 6% incorrect, and the lowest missed- match rate (25%). More importantly, the efficiency gains were realised not in raw computational time but in the speed with which reviewers could process, validate, and finalise matches. Since the RAG system presented each ambiguous match with supporting rationale and a confidence score, the decision to keep or reject a match was straightforward, allowing the review panel to process benchmarking results far more quickly than traditional manual reconciliation.

The use of RAG also proved particularly effective in handling cases where task descriptions were ambiguous, fragmented, or inconsistent across assets. For example, Table 1 highlights tasks such as "arc flash equipment maintenance" and "battery replacement local break station," where different builds used varying descriptions and intervals. The system's ability to retrieve contextual evidence and propose semantic equivalences allowed reviewers to consolidate these into consistent benchmarks. This capability directly supports the industry objective of identifying best practices, harmonising maintenance strategies, and enabling more reliable cross-asset comparisons.

Overall, the results confirm that the integration of RAG represents a substantial advancement over both manual and traditional NLP-based approaches. The combination of automated mapping, human-in-the-loop validation, and audit-ready governance ensures both efficiency and trustworthiness. The modest increase in computational runtime compared to embedding-only methods is offset by vastly improved accuracy, traceability, and decision-making speed, establishing this approach as a scalable and reliable solution for maintenance benchmarking in the upstream oil and gas sector.

## Challenges and Limitations

Despite the demonstrable accuracy and efficiency gains delivered by the RAG workflow, several technical and organisational constraints remain. Acknowledging these boundaries is critical to setting realistic expectations for adoption, managing risk, and identifying areas for ongoing improvement.

### Data Quality, Completeness, and Drift

The effectiveness of RAG is inherently dependent on the quality of underlying CMMS records. Semantic matching is significantly weakened when source tasks are underspecified, such as entries labelled only as "EQ-001 routine check" or "equipment maintenance." In such cases, accuracy drops to ~78%. Similarly, rare equipment classes with fewer than five historical entries average ~72% accuracy due to insufficient embedding representation. Data drift also emerges as a risk, as intervals, codes, and descriptions evolve over time. Without proactive governance, embedding performance deteriorates.

### Language and Terminology Variation

The current embedding models perform strongly on English records, achieving ~94% accuracy. However, tasks expressed in Norwegian, Dutch, or mixed-language formats experience an average 8% decline in accuracy. Morphological and compound-word challenges, for example Norwegian vibrasjonsmåling versus English "vibration check," expose the limitations of general-purpose multilingual embeddings. Although multilingual foundation models help bridge this gap, domain-specific glossaries and per-language threshold calibration remain essential for reliable cross-language benchmarking.

### Generic or Underspecified Tasks

Legacy data migrations and inconsistent entry practices often strip context from task descriptions, producing underspecified records that resist accurate benchmarking. Even with RAG retrieval, the generator frequently returns "no match" or low- confidence outputs, requiring subject-matter experts (SMEs) to intervene. This problem is structural rather than algorithmic: until mandatory descriptive fields or structured task templates are enforced within CMMS systems, ambiguity will persist and remain a barrier to full automation.

### Human-in-the-Loop Dependency

While RAG substantially reduces SME workload, particularly in non-critical cases, complete autonomy is neither achievable nor desirable. All GPT-generated matches continue to demand human validation, especially for safety-critical or conflicting tasks. Moreover, when rationales draw on lengthy or ambiguous passages, review effort increases, partially offsetting efficiency gains. To address this, governance models must incorporate confidence-band routing, rejection pathways, and sampled audits. Provenance mechanisms —embedding versioning, index snapshot hashes, prompt identifiers, and threshold records—remain essential to satisfy auditors and ensure traceability under regulatory scrutiny.

### Geographic and Regulatory Generalisation

The quantitative results reported here are derived from assets in the North Sea and Gulf of Mexico. Applying the workflow to regions with fundamentally different duty cycles, climates, or regulatory regimes—such as Middle Eastern onshore desert operations—will require localised corpus augmentation and threshold recalibration. Without this, transfer accuracy may decline, reducing trust in early deployments and creating resistance to adoption.

**Summary**

RAG offers a step-change in benchmarking efficiency and accuracy, yet it is not a "set-and-forget" solution. Performance depends on disciplined data governance, multilingual adaptation, scheduled re-indexing, and a calibrated human-in-the-loop. Addressing these constraints will convert today's promising pilot results into a resilient, enterprise-wide capability that withstands evolving data, regulatory scrutiny, and operational complexity.

## Future Work and Conclusions

This This study demonstrated that a Retrieval-Augmented Generation (RAG) approach can materially improve the accuracy and efficiency of maintenance benchmarking across heterogeneous CMMS environments. At the task level, accuracy increased from 87% to 94% while analyst time dropped from 10 to 1 hour per 1,000 tasks. These gains translated into higher build alignment (89%), greater uptake of interval recommendations after SME review (78%), and indicative operational benefits, including 12–18% cost savings and a 15% reduction in unplanned shutdowns. Together, the results show that dense semantic retrieval, coupled with grounded generation and governed by human-in-the-loop quality gates, can transform benchmarking from a manual, episodic activity into a repeatable, auditable, and timely practice.

**Next Steps**

Future work should focus on:

- **Evaluation**: adopting operator-level cross-validation and train/validation/test splits to quantify generalisation across sites and geographies.
- **Multilingual adaptation**: addressing the 8% accuracy drop in non-English records through fine-tuned embeddings, bilingual glossaries, and per-language thresholds.
- **Taxonomy and governance**: enforcing structured CMMS templates and aligning to ISO-style equipment taxonomies to lift baseline signal quality.
- **Threshold calibration**: moving from static cut-offs to adaptive, class-specific optimisation to stabilise acceptance rates.
- **Feedback loops**: using SME adjudications as training signals and curating "hard cases" for retraining, reducing the review load.
- **Explainability and auditability**: standardising provenance artifacts (e.g., corpus snapshot IDs, embedding versions, prompt hashes) for compliance and MOC audits.
- **Operational integration**: embedding benchmarking into enterprise workflows through APIs, write-back of accepted matches, and dashboards showing confidence and impact.
- **Scalability**: managing compute via batching, caching, and quantisation, while tracking runtime cost per 1,000 tasks to safeguard efficiency gains.
- **Prospective validation**: moving beyond retrospective exercises to controlled A/B trials measuring reductions in corrective work, downtime, and costs.

**Broader Implications**

RAG-based benchmarking enables more frequent and defensible cross-operator comparisons, supporting reliability through evidence-based intervals, cost discipline via reduced interventions and analyst hours, and compliance through transparent, traceable recommendations. Crucially, it also captures and compounds SME knowledge over time, strengthening institutional memory. While full autonomy is neither realistic nor desirable for safety-critical contexts, the combination of dense retrieval, grounded generation, and calibrated human review provides a scalable path forward. With improved multilingual support, stronger data governance, and prospective validation, RAG can evolve into a standard industry tool for maintenance strategy development, accelerating learning cycles and raising the baseline of reliability performance across the energy sector.

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